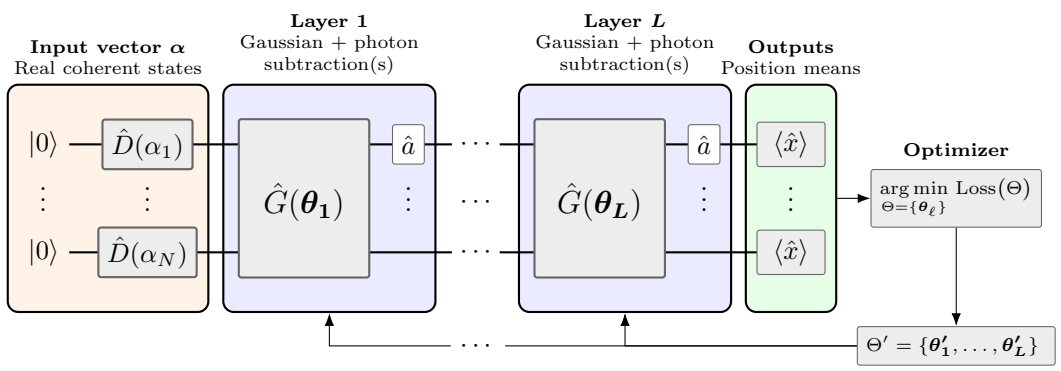

Input vector α
Real coherent states
Layer 1
Gaussian + photon subtraction(s)
Layer L
Gaussian + photon subtraction(s)
Outputs
Position means
$|0\rangle$
$|0\rangle$
$\hat{D}(\alpha_1)$
$\hat{D}(\alpha_N)$
$\hat{G}(\boldsymbol{\theta_1})$
$\hat{G}(\boldsymbol{\theta_L})$
$\hat{a}$
$\hat{a}$
$\langle\hat{x}\rangle$
$\langle\hat{x}\rangle$
Optimizer
$\arg\min_{\Theta=\{\boldsymbol{\theta}_\ell\}} \mathrm{Loss}(\Theta)$
$\Theta' = \{\boldsymbol{\theta'_1}, \ldots, \boldsymbol{\theta'_L}\}$